Zhiyu Wu

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EDUCATION

University of Illinois Urbana-Champaign **Champaign, IL**
M.S. Computer Science – Thesis Based GPA: 3.78 / 4.00

University of Michigan **Ann Arbor, MI**
B.S.E. Computer Engineering (Dual Degree) *August 2022 – May 2024*

Shanghai Jiao Tong University **Shanghai, China**
B.S.E. Electrical and Computer Engineering (Dual Degree) GPA: 3.75 / 4.00 *Sept. 2020 – August 2024*

Research Interest: LLMs, ML system infrastructure, systems for LLMs
Coursework: Computer Architecture, Operating Systems, Computer Network, Machine Learning, Compiler, Algorithms

WORK EXPERIENCE

Research Engineering Intern, Bytedance, Data-AML-MLSys group **San Jose, CA**
Supervisor: Xiao Yu *May 2025 –August 2025*

- Built end-to-end AI model training workflow with Jupyter notebook and web frontend, encompassing dataset profiling, training parameter optimization, SFT/RL deployment, evaluation, and production inference pipeline.
- Contributed to LLM training simulator with memory analysis, communication modeling, and MFU metrics.
- Explore and optimize memory usage for SFT/pretrain with FSDP, specifically, optimize compiler/kernel to maximum tensor reuse in backward and optimizer stage.

RESEARCH EXPERIENCE

JVM for LLM Agentic Systems **Champaign, IL**
Supervisor: Fan Lai *October 2025 - present*

- Designed an ML Agent Compiler and Fork Engine to address reliability bottlenecks in LLM-based agent systems, enabling robust state management for multi-step, stateful tasks.
- Implemented a runtime environment with dual-graph tracking (Agent Workflow and System State) to manage complex coordination and parallel exploration.
- Developed multi-layer fault tolerance mechanisms, including attribute propagation, checkpointing via fork/merge, and liveness counters for resource deallocation.
- Optimized execution efficiency with "Lookahead Compilation" to predict failure-prone branches and reduce checkpointing overhead.
- Introduced RDD-style deduplication to cache and reuse partial execution results, significantly reducing recovery costs and latency in stochastic environments.

Cacheflow: Cross-layer KV Cache Parallelism for LLM Serving at Scale **Champaign, IL**
Supervisor: Fan Lai *August 2024 - present*

- Identified KV cache as a growing memory and latency bottleneck in large-scale LLM serving systems.
- Optimized TTFT by jointly leveraging prefills and KV-cache reuse, introducing token-wise, layer-wise, and multi-GPU parallel restoration strategies tailored to different hardware configurations.
- Designed cross-layer management system that optimally balances computation and communication for KV cache restoration at cluster scale.

Tempo: Application-aware LLM Serving with Mixed SLO Requirements **Champaign, IL**
Supervisor: Fan Lai *July 2024 – Sept. 2025*

- Identified diverse SLO requirements across three LLM request patterns: latency-sensitive, throughput-sensitive, and compound requests with varying performance needs.
- Developed SLO-aware scheduler that maximizes service gain by allocating just enough bandwidth to meet individual SLOs while preserving residual capacity for other requests.
- Constructed multi agent workflow as a DAG, with input and output length as weights, and apply graph matching to predict dependency and remaining length for compound requests.

- Implemented quantile regression forests for response length estimation.
- Created service density-based prioritization algorithm that balances requests across different SLO constraints without starving low-priority workloads.
- Achieved 1.3×-8.3× improvement in service gain and 4.0×-10.3× improvement in SLO goodput compared to existing systems across diverse workloads and models.

Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services **Ann Arbor, MI**
Supervisor: Mosharaf Chowdhury *May 2023 – April 2024*

- Identified that in LLM text-streaming services, systems must generate faster than user reading speed to enhance user experience, addressing gaps in previous metrics.
- Defined Quality of Experience (QoE) in LLM serving by tracking each step of text generation and monitoring the overall user experience throughout the entire streaming process.
- Formulated the problem as a knapsack optimization and developed a scheduling algorithm to maximize QoE in online LLM serving.
- Built Andes, an LLM serving system on top of vLLM, integrating the scheduling algorithm to enhance QoE in real-time LLM services.
- Co-authored the paper "Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services" as the second author.

PUBLICATIONS

- Tempo: Application-aware LLM Serving with Mixed SLO Requirements; Preprint, 2025; Wei Zhang*, **Zhiyu Wu***, Yi Mu, Banruo Liu, Myungjin Lee, Fan Lai
- Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services; Preprint, 2024; Jiachen Liu, **Zhiyu Wu**, Jae-Won Chung, Fan Lai, Myungjin Lee, Mosharaf Chowdhury
- The ML. ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization; NeurIPS, spotlight, 2025; Jae-Won Chung, Jiachen Liu, Jeff J Ma, Ruofan Wu, Oh Jun Kweon, Yuxuan Xia, **Zhiyu Wu**, Mosharaf Chowdhury

PROJECT EXPERIENCE

Symbiotic Lab/ML.ENERGY.LEADERBOARD Team **Ann Arbor, MI**
Developer *May 2023 – Sept. 2023*

- Developed the ML.ENERGY Leaderboard, an open-source platform for benchmarking the energy efficiency and NLP performance of LLM models.
- Defined performance metrics and implemented scripts for optimized batched inference to ensure accurate measurement.
- Contributed to the online Chatbot Arena which gathers data on models' energy consumption and performance.

Embedded Device for Keystroke Timing and Acoustic Attack Protection **Ann Arbor, MI**

- Designed the device to intercept keystrokes and introduce random delays before sending to the PC.
- Implemented keystroke sound playback to counter acoustic attacks using recorded sounds.
- Based the system on the STM32F405 microcontroller with Embedded Rust for secure and efficient performance.
- Delivered a compact, user-friendly design with production costs around $24.50 per unit.
- Utilized SD card for storing custom keystroke sounds and USB peripherals for communication with keyboard and host PC.

PROFESSIONAL SERVICE

- VP 160 Honors Physics SJTU, 2022 Summer

SKILLS

Computer: C++, C, Python, Rust, Java, Pytorch, CUDA, System Verilog, Embedded C/Rust, Linux, MATLAB, Git, LaTeX, LLVM, JAX

HONORS

Dean List, *Umich*	2023
University Honor, *Umich*	*2022*
Tang Junyuan Scholarship, *SJTU*	*2022*
SJTU Undergraduate Excellent Scholarship Class B, *SJTU*	*2022*